

TrES-2b

R-2068

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_120527 · created 2026-08-09T22:57:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456074.95101 +/- 0.00015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.076 +/- 0.037
Transit depth (Rp/Rs)^2	0.58 +/- 0.56 [%]
Orbital Inclination (inc)	84.16 +/- 2.76
Airmass coefficient 1 (a1)	1.613 +/- 0.015
Airmass coefficient 2 (a2)	-0.45 +/- 0.2
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	#1 - [505, 431]
Optimal Method	PSF photometry
Transit Duration (day)	0.047 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.076 ± 0.037 vs 0.1254 (deviation 1.34σ)
Duration fit vs published	0.047 d vs 0.0746 d (deviation 0.99σ)
Mid-time O-C	-0.2 min
Depth SNR	2.1
Residual scatter	0.94 %

Light curve

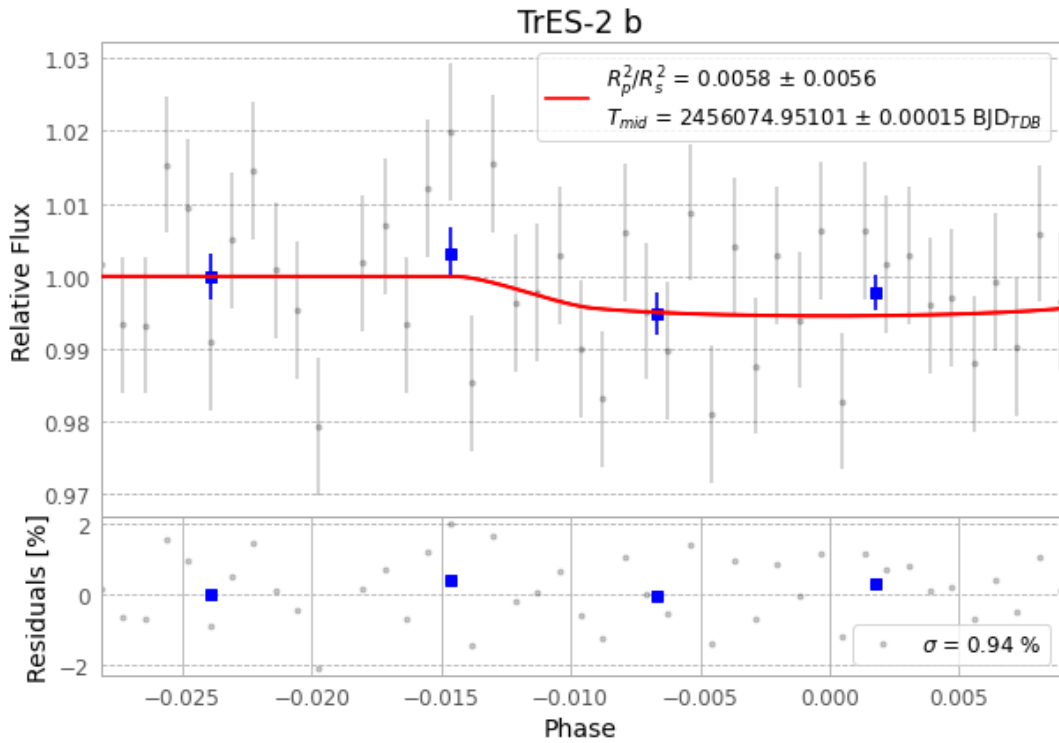


Figure 1 — FinalLightCurve_TrES-2 b_2012-05-27.png (SHA-256 c43cb68bf7d7d34e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [444, 273], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 025f4e45db9ac6f5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

45 FITS frames (30 MB), TRES-2120527090614.FITS → TRES-2120527111813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-120527.log (53fe388817da...), FinalLightCurve_TrES-2 b_2012-05-27.png (c43cb68bf7d7...), FinalLightCurve_TrES-2 b_2012-05-27.csv (0e08320ebddf...)

Compute resources

Wall time 496.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 22:57Z.